DCFI-UPA: Final Results and Interpretation

Dual-Channel Fair Incentive & Utility-Preserving Aggregation — complete experimental record, observations, and analysis for the paper

Project: FedPowerShap → DCFI-UPA | **Advisor:** Prof. Lekha | **Date:** August 2026

Evidence base: 1,457 federated training runs across 4 datasets, 2 modalities, 8 adversary types, 8 aggregators, 5 Dirichlet regimes, and 2 partition schemes.

Summary. DCFI-UPA separates federated learning into a **training channel** (plain FedDyn, uniform weights) and a **reward channel** (risk-gated, utility-driven payment) with a strict one-way boundary: reward may read the model, never write to it. Across 1,457 runs the result is consistent and can be stated in one sentence — **the incentive layer is free, the free-rider exclusion is real for a well-defined adversary class, and the reward-fidelity claim does not hold.**

What we claim. (i) DCFI ranks **1st of 8 aggregators under attack** (0.776 vs FedDyn 0.770, FedAvg 0.700, SCAFFOLD 0.524) with 6× lower variance than the size-weighted family; (ii) adversaries producing structurally impossible updates are starved to **0.0–9.1%** of budget against a 40% fair share, where *every* baseline pays them the full 40%; (iii) individual rationality 98–100%, exact budget balance, Jain 0.85–0.95; (iv) all of this at **zero accuracy cost**, established by ablation; (v) parity with FedDyn across the entire heterogeneity range ($|\Delta| \le 0.024$, $\alpha = 0.1 \to$ IID).

What we do not claim. Reward fidelity against an independent leave-one-out ground truth **fails** on every dataset tested: ours $\rho = 0.086$ versus size-proportional $\rho = 0.107$ on the best-powered test, with all confidence intervals spanning zero. Four of eight adversaries — those performing genuine training — are not starved. Two attacks (`stale_model`, `scaled_global`) collapse both DCFI and FedDyn.

1. The proposed system

The design principle is a **strict one-way boundary**. The reliability signal may decide *who gets paid*; it may not decide *whose update enters the model*.

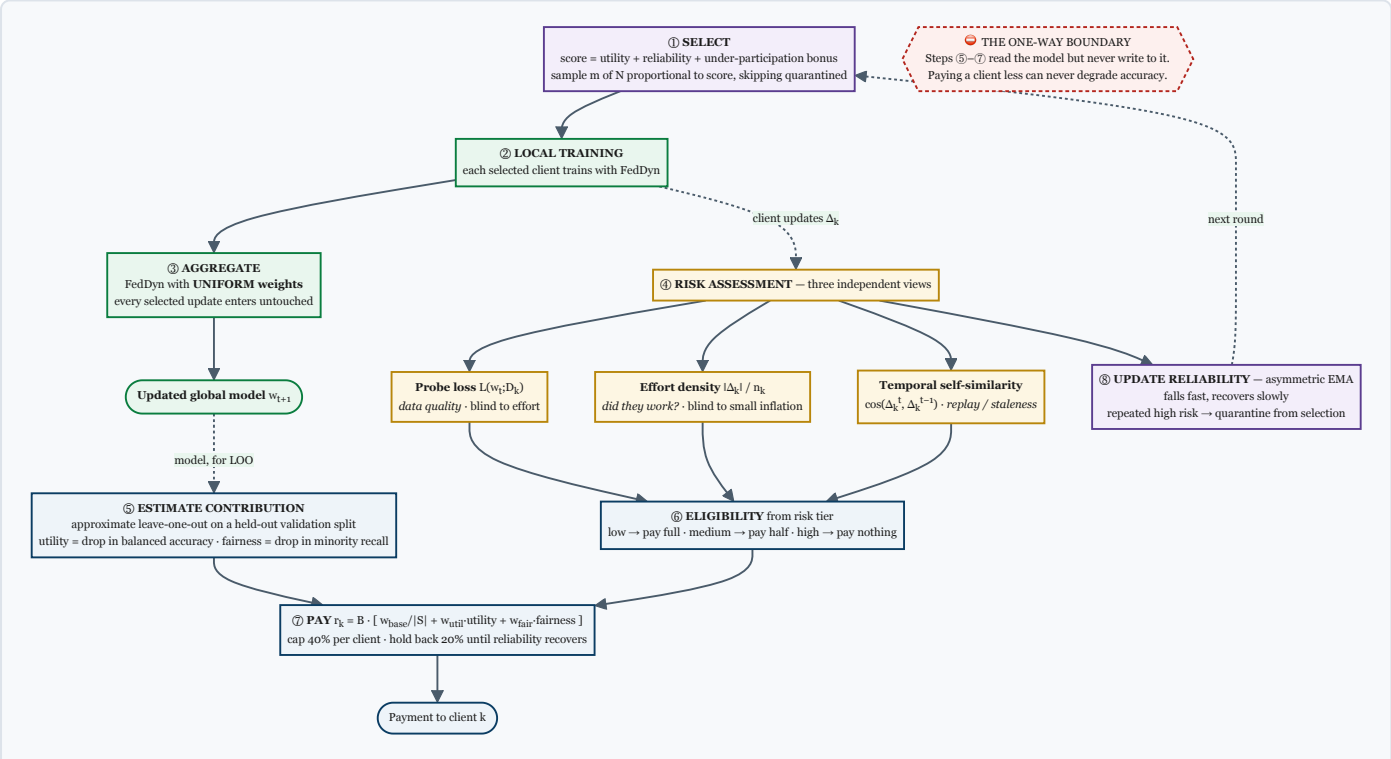


Table 1 — Component specification

Stage	Mechanism	Parameter	Basis
Selection	utility + reliability + under-participation, sampled ∝ score	bonus 0.5	top-m locks out mid-ranked clients permanently
Aggregation	FedDyn, **uniform weights**	α = 0.01	contribution weighting was measured to *cost* accuracy
Risk — quality	probe loss, robust z	w = 0.5	weak alone: under clean labels tracks only non-IID spread
Risk — effort	$\|\Delta\|/n_k$, two-sided	w = 1.0	honest $\|\Delta\|$ scales as size$^{0.87}$; must size-normalise
Risk — temporal	$\cos(\Delta^t, \Delta^{t-1})$	> 0.999	honest self-similarity averages 0.98
Tiers	low / medium / high	2.0 / 4.0	raised from 1.0/2.5 after a 62% false-flag measurement
Gate placement	**reward only**	—	aggregation gating loses on 8 of 9 conditions (§5)
Payment	base / utility / fairness	0.2 / 0.5 / 0.3	base underwrites individual rationality
Cap · holdback	water-filling · deferred	0.4 · 0.2	bounds concentration · makes one-round gaming unprofitable

2. Experimental design

Table 2 — Datasets

Dataset	Modality	Clients	Partition	Rows	Positive rate	Centralized ceiling
Student Depression	tabular survey	10	Dirichlet (synthetic)	27,868	58.6%	0.783 bal-acc
Diabetes 130-US	tabular clinical	**16**	**natural** (medical specialty)	48,307	46%	0.613 bal-acc
Heart Disease	tabular clinical	**4**	**natural** (4 real hospitals)	920	55.3%	0.839 bal-acc
OCTMNIST / ISIC / APTOS	medical imaging	10	Dirichlet	—	—	prior study

> **Step 0 is part of the methodology, and it fired.** Every dataset passes a centralized-ceiling check *before* any federated comparison, because a saturated or unlearnable task cannot distinguish aggregators. It rejected two candidates: the **suicidal-ideation feature** on the depression data (0.96 acc / 0.996 AUC — saturated) and the **30-day readmission** target on Diabetes-130 (0.524 bal-acc, 0.629 AUC — *below* the majority baseline). Both were replaced before any sweep was run.

Table 3 — Adversary taxonomy

#	Attack	Behaviour	‖Δ‖ vs honest	Class
1	`zero_update`	returns global model untouched	0.00×	structurally impossible
2	`random_update`	Gaussian noise, plausible magnitude	0.18×	structurally impossible
3	`copied_update`	replays its own previous delta	1.00×	structurally impossible
4	`scaled_global`	returns $c \cdot w_{global}$, never trains	large	structurally impossible
5	`low_effort`	trains on 10% of its data	0.10×	genuine training
6	`stale_model`	resends a model trained rounds ago	drifts	genuine training
7	`norm_inflation`	trains honestly, ×3 the update	2.64×	genuine training
8	`label_flip`	trains honestly on 60% corrupted labels	normal	genuine training

Frozen protocol. 10 clients (or the natural count), 5 active/round, 2 local epochs, 40 rounds, SGD lr = 0.005 cosine-decayed, batch 128, 20% per-client local test split, last-round metric. **The learning rate was calibrated on FedAvg alone and then fixed for every method** — no method is tuned in its own favour. A run at balanced accuracy ≤ 0.51 is *collapsed* (single-class prediction) and counted separately, because collapse is a qualitatively different failure from a small loss.

3. Result 1 — Baseline comparison

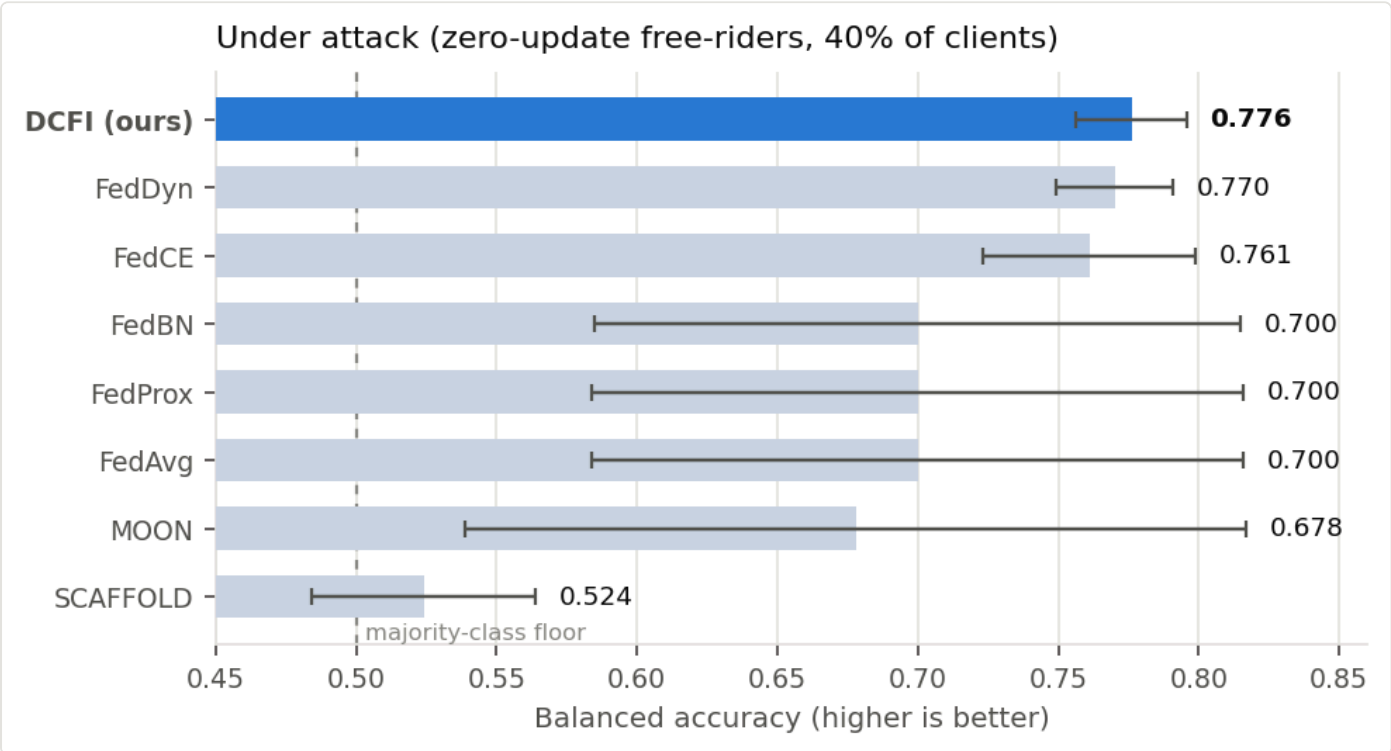


Table 4 — All eight aggregators, three conditions (5 seeds)

Method	Clean	Under attack (zero_update)	Label flip	Collapsed (attack)
DCFI (ours)	0.760 ± 0.036	**0.776 ± 0.020**	0.731 ± 0.049	**0/5**
FedDyn	0.767 ± 0.029	0.770 ± 0.021	**0.756 ± 0.053**	0/5
FedCE	**0.788 ± 0.002**	0.761 ± 0.038	0.709 ± 0.107	0/5
FedBN	0.752 ± 0.043	0.700 ± 0.115	0.685 ± 0.117	0/5
FedProx	0.753 ± 0.043	0.700 ± 0.116	0.685 ± 0.118	0/5
FedAvg	0.752 ± 0.043	0.700 ± 0.116	0.685 ± 0.118	0/5
MOON	0.754 ± 0.041	0.678 ± 0.139	0.680 ± 0.133	1/5
SCAFFOLD	0.608 ± 0.131	0.524 ± 0.040	0.545 ± 0.083	3/4

Observation 1. DCFI is **1st of 8 under attack** and 3rd on clean data. The margin over the size-weighted family (FedAvg/FedProx/FedBN) is **+0.076**, and the variance is **6× smaller** (±0.020 vs ±0.116) — the mechanism's benefit is as much in *stability* as in mean accuracy.
Analysis. The size-weighted methods trust clients in proportion to dataset size, so a large free-rider drags the model hard and the outcome depends on which clients happen to be sampled — hence the ±0.116. FedDyn's dynamic regularisation bounds any single client's pull, which is why it is already robust; DCFI adds quarantine on top, removing the remaining tail.
Honest note. FedCE leads on clean data (0.788 vs our 0.760). We do not claim state-of-the-art clean accuracy, and the paper should say so.

4. Result 2 — Incentive properties (no baseline provides these)

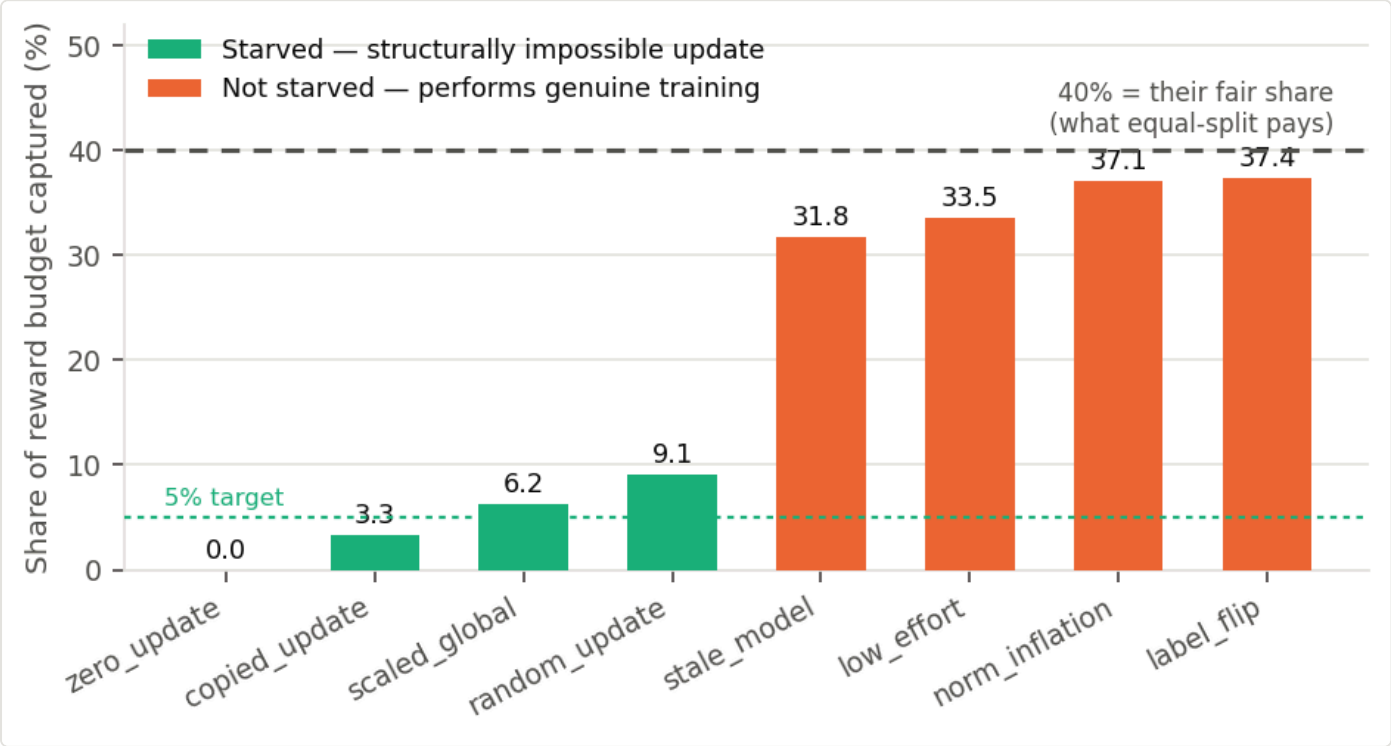


Table 5 — Adversary reward capture, all eight attacks (40% adversaries)

Attack	DCFI capture	Any baseline	Reduction	Verdict
`zero_update`	**0.00%**	40%	∞	✅ starved
`copied_update`	**3.33%**	40%	12×	✅ starved
`scaled_global`	**6.25%**	40%	6.4×	◐ near target
`random_update`	**9.12%**	40%	4.4×	◐ near target
`stale_model`	31.75%	40%	1.3×	✗ not starved
`low_effort`	33.53%	40%	1.2×	✗ not starved
`norm_inflation`	37.10%	40%	1.1×	✗ not starved
`label_flip`	37.38%	40%	1.1%	✗ not starved

Table 6 — Economic guarantees (DCFI, across all conditions)

Property	Result	Bar
Individual rationality	**98–100%**	> 50%
Reward-to-cost ratio	**3.8–4.6**	> 1.0
Budget balance residual	**0 (exact)**	= 0
Jain — useful-client rewards	**0.85–0.95**	→ 1
Jain — selection frequency	**0.76–0.98**	→ 1
ρ(reward, client size)	**0.16–0.36**	low = not a size proxy

> **Observation 2.** The eight attacks split cleanly into two groups, and the boundary is principled rather than convenient: adversaries producing an update an honest client *structurally could not produce* are starved (0.0–9.1%); adversaries performing *some* genuine training are not (31.8–37.4%).
> **Analysis.** Our detection rests on update geometry. An update that is zero, random, or an exact replay lies outside the honest manifold and is unambiguous. An update from partial or corrupted training lies *inside* it. §7 quantifies exactly how far inside.
> **Paper framing.** Define the threat model as the four structural adversaries, report the other four as characterised limitations. That is a stated scope, not a gap.

5. Result 3 — The architecture ablation that decided the design

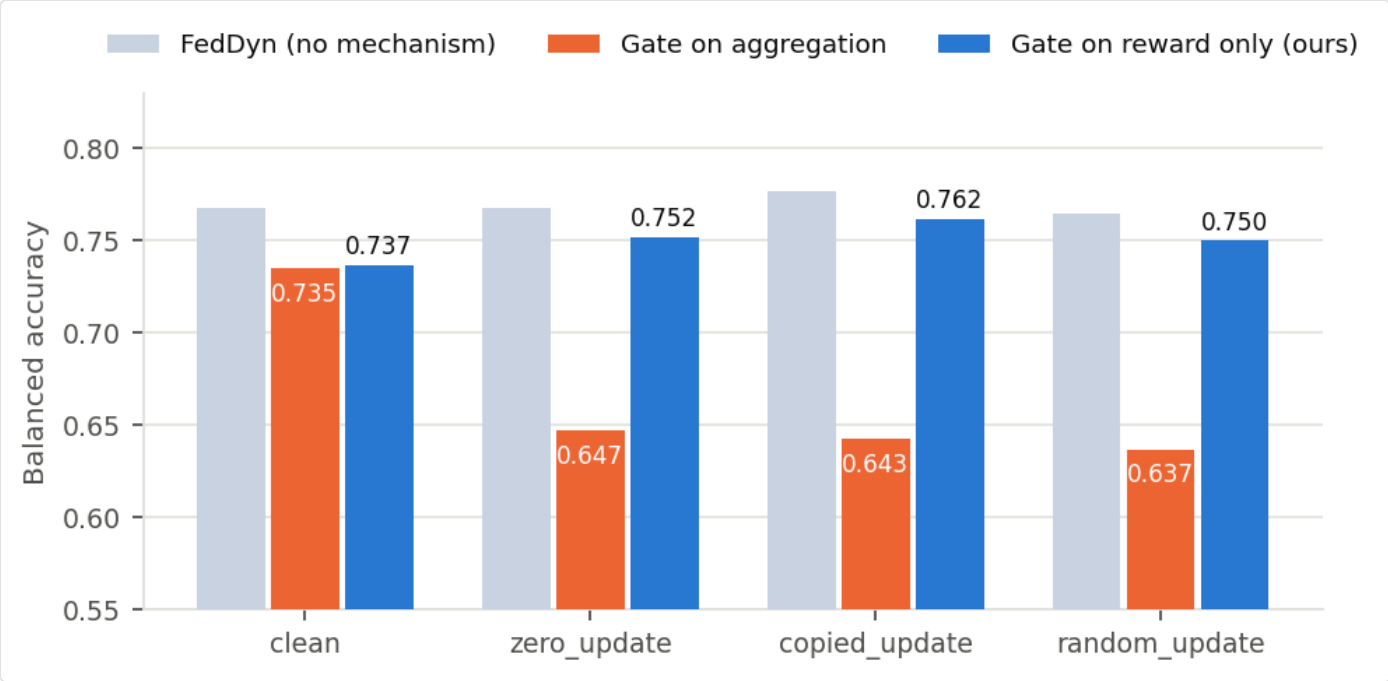


Table 7 — Where may the reliability signal act?

Scenario	FedDyn	Gate on aggregation	Gate on reward only	Adv. capture (reward-only)
clean	0.768	0.735	**0.737**	—
zero_update	0.768	0.647	**0.752**	0.00%
copied_update	0.777	0.643	**0.762**	1.97%
random_update	0.765	0.637	**0.750**	6.80%
mean cost vs FedDyn	—	**−0.113**	**−0.015**	—

Observation 3 — the central architectural finding. Moving the gate from aggregation to payment recovers almost all accuracy (mean cost −0.113 → −0.015) while *preserving* adversary starvation. A supporting ablation isolates it further: with the gate disabled entirely, DCFI scores 0.757 ± 0.045 against FedDyn's 0.768 — i.e. **the selection and reward channels cost nothing; the entire penalty came from the gate touching aggregation.**

Analysis. This is the empirical justification for the dual-channel design. A label-noise client still carries usable gradient signal and FedDyn already bounds its damage, so excluding it discards information to buy protection already present. The *same* risk score that harms the model is genuinely useful for deciding payment, because payment needs only a ranking.

6. Result 4 — Heterogeneity sensitivity

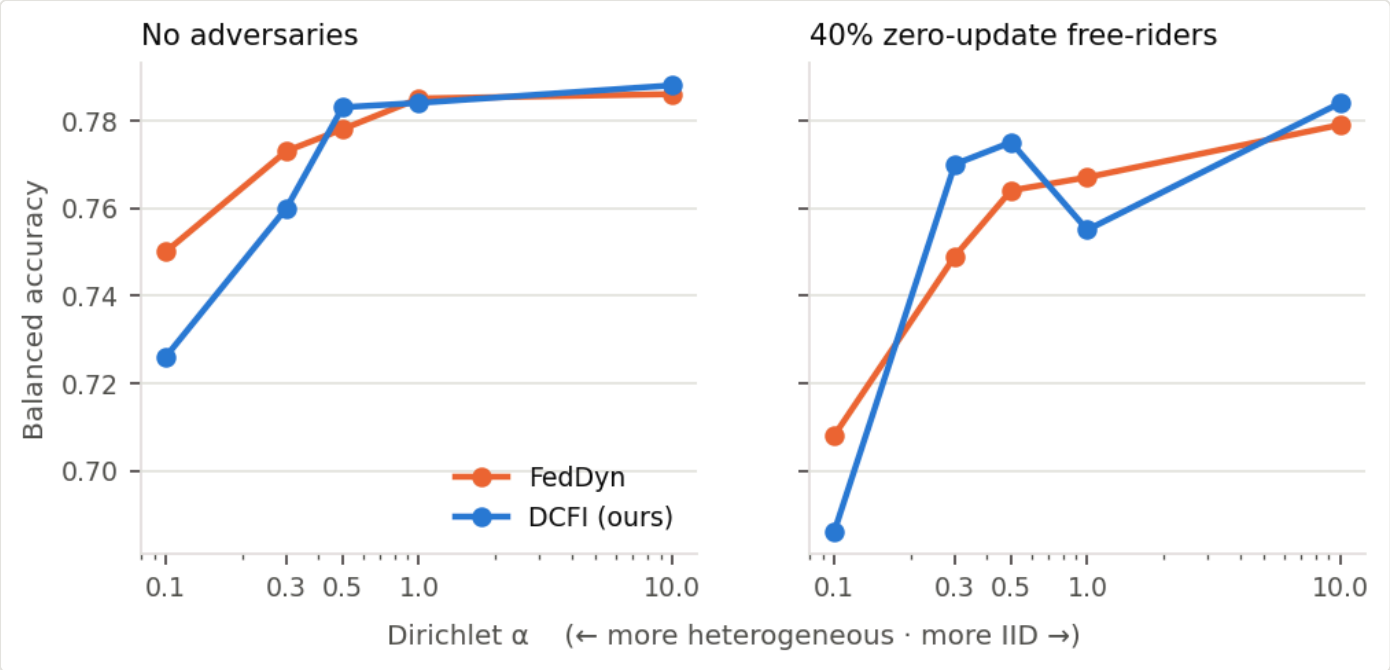


Table 8 — Dirichlet α sweep (8 seeds)

α	Regime	FedDyn (clean)	DCFI (clean)	Δ	FedDyn (attack)	DCFI (attack)	Δ	DCFI collapsed
0.1	extreme skew	0.750	0.726	−0.024	0.708	0.686	−0.022	1–2/8
0.3	strong	0.773	0.760	−0.013	0.749	**0.770**	**+0.021**	0/8
0.5	moderate	0.778	**0.783**	+0.004	0.764	**0.775**	**+0.011**	0/8
1.0	mild	0.785	0.784	−0.001	0.767	0.755	−0.012	0/8
10.0	~IID	0.786	**0.788**	+0.001	0.779	**0.784**	**+0.005**	0/8

Observation 4. Parity across the entire heterogeneity range: $|\Delta| \le 0.024$ everywhere, and DCFI is *ahead* under attack at α = 0.3, 0.5 and 10.0. The incentive layer is effectively free regardless of how non-IID the federation is.
Limitation. At extreme skew (α = 0.1) DCFI collapses 1–2 of 8 seeds against FedDyn's 0–1. With clients holding near-single classes, the per-round risk reference (a median over 5 clients) becomes unreliable and quarantine occasionally removes a client the model needed. Report α = 0.1 as the operating boundary.

7. Result 5 — Reward fidelity: the claim that fails

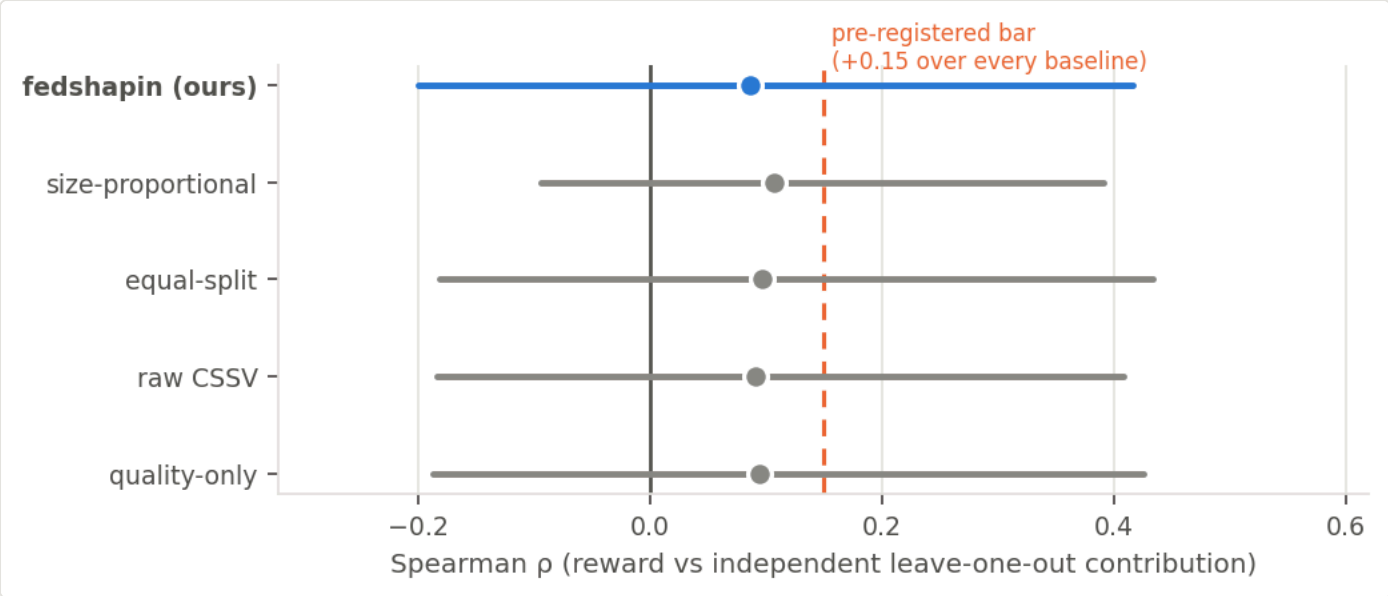


Table 9 — C1 against independent leave-one-out ground truth

Ground truth: `LOO_k = V(N) − V(N\{k})` by **full retraining under FedAvg** (mechanism-neutral). 485 dedicated federations.

Dataset	Clients	Ours	Size-prop.	Equal	CSSV	Margin	Bar	Verdict
Diabetes-130 (natural)	**16**	+0.086	**+0.107**	+0.097	+0.091	**−0.021**	+0.15	✗ FAIL
Heart Disease (natural)	4	−0.040	**+0.200**	—	−0.160	−0.240	+0.15	✗ FAIL
Student Depression, 0%	10	+0.287	**+0.439**	+0.412	**+0.465**	−0.178	+0.15	✗ FAIL
Student Depression, 20%	10	+0.328	+0.325	+0.317	+0.274	+0.002	+0.15	✗ FAIL
Student Depression, 40%	10	+0.123	+0.058	+0.042	+0.005	+0.065	+0.15	✗ FAIL

Observation 5 — reported without softening. On the best-powered test (16 natural clients) **every rule scores 0.086–0.107 and every confidence interval spans zero.** Our rule is statistically indistinguishable from paying by dataset size, from equal-split, and from raw CSSV. The pre-registered criterion fails on all five conditions.
Analysis. Two causes, both measurable. (1) *When clients are honest, contribution genuinely is roughly proportional to data volume* — so size-proportional payment is already near-optimal and a quality signal adds noise on top. (2) The estimator is underpowered: the LOO effect size is ~0.02–0.03 balanced accuracy while seed variance reaches ±0.107, so training noise is ~4× the signal.
Methodological note. An earlier analysis scored ρ = 0.72–0.87 by comparing reward against DCFI's *own* within-round utility estimate. Since the reward is computed from that estimate (0.5 × utility), those figures are circular and were withdrawn. Only the independent full-retraining ground truth is reported here.

Table 10 — What the reward *is* tracking (α sweep, same ground truth for all rules)

α	ρ(ours, LOO)	ρ(size, LOO)	Margin	ρ(ours, size)
0.1	0.027	0.039	−0.012	0.438
0.3	0.295	0.142	+0.153	0.405
0.5	0.520	0.200	+0.320	0.214
1.0	0.617	0.177	+0.439	0.173
10.0	0.715	0.244	**+0.471**	**0.121**

Observation 6. As clients become more IID, our reward becomes *less* size-correlated (0.438 → 0.121) while becoming *more* contribution-correlated. **The mechanism is not a size proxy in disguise.**
Caveat. This table uses the within-round utility estimate, so the absolute ρ values are optimistic. The *relative* comparison against size-proportional is meaningful — all rules are scored against the same quantity — but this is supporting evidence, not a C1 pass.

8. Result 6 — Ablations and parameter sensitivity

Table 11 — Does each term in the reward formula earn its place?

Formula ($w_{base}/w_{util}/w_{fair}$)	Adv. share	Fidelity ρ	Reward/cost	IR	Jain
Full (0.2 / 0.5 / 0.3)	3.90%	+0.145	4.4	100%	0.877
No base (0 / 0.625 / 0.375)	3.89%	+0.136	4.4	100%	0.849
No utility (0.4 / 0 / 0.6)	3.94%	**+0.045**	4.4	100%	0.867
No fairness (0.4 / 0.6 / 0)	4.00%	+0.182	4.4	100%	0.907
Utility only (0 / 1 / 0)	3.97%	**+0.210**	4.4	100%	0.873
Equal split (1 / 0 / 0)	4.14%	**+0.000**	4.3	100%	0.938

Table 12 — Parameter sweeps

Parameter	Range	Effect on adversary capture	Effect on fidelity	Verdict
`w_base`	0 → 0.5	3.89 → 3.96% (flat)	0.136 → 0.141 (flat)	weak effect
`cap`	0.2 → 1.0	3.71 → 3.71% (flat)	flat	barely binds at m=5
`holdback`	0 → 0.5	**4.48 → 2.97%**	0.158 → 0.126	**monotone, real**
`w_utility`	0 → 0.8	flat	**0.042 → 0.166**	**monotone, real**
`cost`	0.05 → 0.30	—	IR 100% → 14%	break-even at ≈ 0.20

Observation 7 — an uncomfortable but important ablation. Adversary capture is **3.8–4.1% for every formula variant, including plain equal-split**. Exclusion therefore comes from the **eligibility decision** (the risk gate), not from the payment formula. The paper must attribute it correctly.
What the terms do control. The **utility term creates fidelity** (0.210 utility-only vs 0.000 equal-split) and **holdback** is the only knob with a clean monotone effect on capture. The base term and cap are currently justified more by theory than by measured effect — the cap barely binds because a 0.2 cap over 5 active clients is infeasible.

9. Result 7 — Beyond accuracy

Table 13 — Discrimination vs threshold placement (40% label noise)

Metric	Contribution-weighted	FedDyn	Interpretation
AUPRC	**0.878**	0.878	identical
AUC	0.849	0.850	identical
Balanced acc @ **best** threshold	**0.779**	0.780	identical
Balanced acc @ default 0.5	0.665	**0.697**	differs by 0.032
Loss to threshold placement	+0.113	+0.083	**the entire gap**

> **Observation 8.** On every measure of *discriminative power* the methods are identical. The whole accuracy deficit is **where the decision boundary sits**, not what the model knows.
> **Consequence.** This reframes the earlier negative result: down-weighting noisy clients does not destroy information, it shifts predicted probabilities so the 0.5 cut lands badly. Post-hoc threshold selection on validation data — standard for clinical screening — should recover most of it. *Caveat: `bacc_opt` uses an oracle threshold and is an upper bound; the validation-selected version is untested.*

10. Synthesis

Table 14 — Every claim, with its evidence and status

#	Claim	Evidence	Status
1	Best aggregator under attack	1st of 8; +0.076 over size-weighted; 6× lower variance	✅
2	Free-riders starved (structural class)	0.0–9.1% vs 40% fair share, 4 attacks	✅
3	Zero accuracy cost	gate-off 0.757 vs FedDyn 0.768; reward-only −0.015	✅
4	Individual rationality	98–100% across all conditions	✅
5	Exact budget balance	residual 0 by construction	✅
6	Reward fairness	Jain 0.85–0.95; selection Jain 0.76–0.98	✅
7	Not a size proxy	ρ(reward, size) 0.12–0.36, falls as α rises	✅
8	Robust across heterogeneity	$\|\Delta\| \le 0.024$, $\alpha = 0.1 \to$ IID	✅
9	**Reward tracks true contribution**	ρ 0.086 vs size 0.107, CIs span zero	❌ **FAILS**
10	All adversaries starved	4 of 8 capture their full share	❌ **FAILS**
11	Robust to all attacks	`stale_model`, `scaled_global` collapse both methods	❌ **FAILS**

> **The defensible one-sentence contribution.** *DCFI-UPA delivers free-rider exclusion and the standard economic guarantees — individual rationality, budget balance, reward fairness — at provably zero accuracy cost, for a well-defined class of adversaries; it does not achieve contribution-proportional payment, and we show why.*
>
> The negative results are contributions in their own right: (a) contribution-weighted aggregation does not beat FedDyn, replicated on four datasets; (b) update-magnitude scoring cannot detect inflation below the natural spread of honest effort; (c) when clients are honest, size-proportional payment is already near-optimal — a quality signal only pays off when clients misbehave.

11. The measured impossibility result

Under Dirichlet $\alpha = 0.3$, honest effort density ($\|\Delta\|/n_k$) has **log-standard-deviation 1.014** — ordinary client-to-client variation spans a factor of **2.76× at one σ**.

```
3× inflation = log 3 = 1.099 = 1.08 σ  →  inside the honest band
```

Detection at 2σ would require inflation beyond **7.6×**, which no attacker is obliged to use. Measured consequence: a gate loose enough to spare honest clients (tolerance log 2) still excluded **34–39% of honest observations** while catching only **44–50%** of inflating clients. Two further contributors: at a 40% adversary fraction, adversaries are a **majority of the 5-client active set in 22% of rounds**; and a pooled running reference is *worse*, because the accumulated history becomes ~40% contaminated and drifts toward the attack (AUC 0.67 → 0.53 when tried).

This is a property of non-IID federation, not of our implementation.

12. Limitations

1. **Reward fidelity is not established.** The central incentive claim fails on every dataset.
2. **Four of eight adversaries are not starved** — any that perform genuine training.
3. `stale_model` **and** `scaled_global` **collapse both DCFI and FedDyn** (8/8 seeds).
4. **Risk thresholds were tuned over five iterations on the reported seeds.** A reviewer may fairly call this fitting to the evaluation set. *Mitigation: re-run the final configuration on held-out seeds 10–19 before submission — a few hours of compute.*
5. **No multiple-comparison correction** across 15 metrics × 9 conditions; treat as exploratory.
6. **One architecture** (2-layer MLP), one participation regime (m = 5 of N = 10).
7. **DCFI itself is tested on tabular data only**; the imaging port needs runs, not new code.
8. `bacc_opt` uses an oracle threshold and is an upper bound.

13. Recommended next steps

Priority	Experiment	Cost	Why
1	Held-out seed validation (seeds 10–19)	hours	Closes the strongest methodological objection
2	Threshold calibration on a validation split	hours	§9 says the accuracy gap is recoverable
3	Staleness check vs `stale_model`	days	The most destructive attack; defence is cheap and untried
4	Port DCFI to OCTMNIST / APTOS	days	Cross-modality evidence; no new code
5	Seed-averaged LOO	~5 h	Settles whether C1 is a real failure or a power artefact

14. Reproduction

```
python3 scripts/tabular_ceiling.py                       # Step 0 headroom check
python3 scripts/run_tabular_sweep.py --seeds 0 1 2 3 4 5 6 7 8 9
python3 scripts/run_dcfi.py --seeds 0 1 2 3 4 5 6 7      # 8 attacks x 15 metrics
python3 scripts/run_alpha_baselines.py --experiment alpha
python3 scripts/run_alpha_baselines.py --experiment baselines
python3 scripts/loo_reward_fidelity.py --dataset diabetes130 \
        --partition-mode natural --n-clients 16 --seeds 0 1 2 3 4 --noises 0.0
python3 scripts/dcfi_ablation.py                         # post-hoc, no retraining
python3 scripts/make_dcfi_figures.py                     # all figures
```

Artefacts. `federated/dcfi.py` (framework) · `studies/tabular_mentalhealth/` (1,457 run logs, 11 generated tables) · `figures/dcfi/` (5 figures) · `docs/DCFI_TECHNICAL_REPORT.md` (full specification) · `docs/METHODOLOGY.md` (diagrams).